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Linear System Theory — Master Notes

EE 545 | Lectures 2–23 | Complete Study Reference

CHAPTER 1: GEOMETRIC VIEW OF ALGEBRA (L02)

Course Philosophy

The central theme: find the geometric picture behind every algebraic expression. An equation may define a hyperplane; a constraint may define a half-space; a dataset may be a point cloud. Linear algebra translates between algebraic and geometric languages.

Vectors

A vector in $\mathbb{R}^n$ is an ordered n -tuple. The order of entries matters. Views:

- **Position:** a point in n -dimensional space.
- **Arrow:** direction and magnitude from the origin.
- **Data container:** e.g., a student represented by (midterm, final, homework).

Non-geometric data (grades, sensor readings, pixel values) becomes geometric by treating it as a point in $\mathbb{R}^n$.

Linear, Affine, and Convex Combinations

Type	Formula	Constraint
Linear	$\sum_{i=1}^m \alpha_i x_i$	None
Affine	$\sum_{i=1}^m \alpha_i x_i$	$\sum \alpha_i = 1$

Type	Formula	Constraint
Convex	$\sum_{i=1}^m \alpha_i x_i$	$\sum \alpha_i = 1, \alpha_i \geq 0$

Span: Set of all linear combinations of $\{x_1, \dots, x_m\}$. Always a subspace.

Affine hull: All affine combinations of a set. Two points $\rightarrow$ the line through them. Three non-collinear points $\rightarrow$ the plane through them.

Convex hull: All convex combinations. Two points $\rightarrow$ line segment. Three non-collinear points $\rightarrow$ triangle.

Convex set: A set S is convex if for any $x, y \in S$ and $t \in [0, 1]$: $tx + (1 - t)y \in S$. Equivalently: all line segments between points in S stay in S .

Expectation as convex combination: $\mathbb{E}[X] = \sum_i p_i x_i$ where $p_i \geq 0, \sum p_i = 1$.

Hyperplanes and Half-Spaces

A **hyperplane** in $\mathbb{R}^n$: $H = \{x : a^T x = b\}, a \neq 0$.

- In 2D: a line. In 3D: a plane. In higher dimensions: a hyperplane.
- Does NOT need to pass through the origin (unless $b = 0$).
- The vector a is the **normal vector** (orthogonal to H).

Half-spaces: $H^+ = \{x : a^T x \geq b\}$ and $H^- = \{x : a^T x \leq b\}$.

Neural network connection: A simple neuron computes $w^T x - \theta$ and fires iff the result is positive $\rightarrow$ it is a half-space classifier. A neuron identifies which side of a hyperplane the input lies on.

CHAPTER 2: NORMS AND INNER PRODUCTS (L03, L19)

Norm Axioms

$\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$ satisfying: 1. $\|x\| \geq 0; \|x\| = 0 \iff x = 0$ (positive definite) 2. $\|\alpha x\| = |\alpha| \|x\|$ (homogeneity) 3. $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality)

Vector p -Norms

$$\|x\|_p = (\sum_k |x_k|^p)^{1/p}, \|x\|_\infty = \max_k |x_k|.$$

Key cases: - ℓ_1 : taxicab norm; promotes **sparsity** in optimization. - ℓ_2 : Euclidean norm; $\|x\|_2 = \sqrt{x^* x}$. - ℓ_∞ : Chebyshev norm; captures worst-case/maximum error.

As p increases, $\|x\|_p$ decreases for fixed x .

Norm balls: ℓ_1 ball = diamond, ℓ_2 ball = circle, ℓ_∞ ball = square (in 2D).

Sparsity and ℓ_1 Minimization

Minimizing $\|x\|_1$ subject to constraints tends to produce sparse solutions. The ℓ_1 ball's corners lie on coordinate axes; constrained optimization often "hits" a corner, zeroing out most components.

Applications: compressed sensing, feature selection, LASSO regression, filter design.

Euclidean Inner Product and Cauchy-Schwarz

Real: $\langle x, y \rangle = x^T y = \sum_k x_k y_k$.

Complex: $\langle x, y \rangle = y^* x = \sum_k \bar{y}_k x_k$ (conjugate the second argument).

Cauchy-Schwarz: $|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2$.

This is essential for defining the **angle** between vectors:

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\|_2 \|y\|_2}.$$

The inequality guarantees $\cos \theta \in [-1, 1]$.

Orthogonality: $x \perp y \iff \langle x, y \rangle = 0$.

Orthonormal set: pairwise orthogonal and each has unit norm.

CHAPTER 3: VECTOR SPACES AND SUBSPACES (L04)

Vector Space (Formal Definition)

Four ingredients: (1) set of vectors V , (2) scalar field $\mathbb{F}$, (3) vector addition, (4) scalar multiplication. Must satisfy 10 axioms (closure, associativity, commutativity, identity elements, inverses, distributivity).

Examples: $\mathbb{R}^n$, $\mathbb{C}^n$, polynomials of degree $\leq n$, continuous functions on $[a, b]$, matrices with zero trace.

Subspace

A nonempty subset $S \subseteq V$ closed under addition and scalar multiplication. Automatically contains the zero vector.

Test: S is a subspace iff for all $x, y \in S$ and scalars α, β : $\alpha x + \beta y \in S$.

Span, Basis, Linear Independence, Dimension

Span: $\text{span}\{x_1, \dots, x_m\}$ = set of all linear combinations. Always a subspace.

Linear independence: $\sum \alpha_i x_i = 0 \implies \alpha_i = 0$ for all i .

Basis: linearly independent set that spans the space. Two characterizations: (a) minimal spanning set, (b) maximal independent set.

Dimension: number of vectors in any basis (well-defined).

CHAPTER 4: $Ax = b$ — EXISTENCE AND UNIQUENESS (L05, L06, L07)

Two Views of Matrix-Vector Multiplication

Column view: $Ax = x_1a_1 + x_2a_2 + \cdots + x_na_n$. Relevant to **existence**.

Row view: $(Ax)_i = r_i^T x$ (inner product of row i with x). Relevant to **uniqueness**.

Four Fundamental Subspaces

For A of size $m \times n$, rank r :

Subspace	Definition	Lives in	Dim
$\mathcal{R}(A)$	span of columns = $\{Ax\}$	Output $\mathbb{C}^m$	r
$\mathcal{R}(A^*)$	span of rows	Input $\mathbb{C}^n$	r
$\mathcal{N}(A)$	$\{x : Ax = 0\}$	Input $\mathbb{C}^n$	$n - r$
$\mathcal{N}(A^*)$	$\{z : A^*z = 0\}$	Output $\mathbb{C}^m$	$m - r$

Orthogonality: $\mathcal{N}(A) \perp \mathcal{R}(A^*)$ (proof: each row of A has zero inner product with null vectors); $\mathcal{N}(A^*) \perp \mathcal{R}(A)$.

Direct sums: $\mathbb{C}^n = \mathcal{R}(A^*) \oplus \mathcal{N}(A)$; $\mathbb{C}^m = \mathcal{R}(A) \oplus \mathcal{N}(A^*)$.

Existence and Uniqueness

Existence: $Ax = b$ has a solution $\iff b \in \mathcal{R}(A)$.

Existence for every b : $\mathcal{R}(A) = \mathbb{C}^m$ (full row rank, requires $n \geq m$) $\iff$ right inverse exists.

Uniqueness: $\mathcal{N}(A) = \{0\}$ (full column rank, requires $m \geq n$) $\iff$ left inverse exists.

All solutions: $\hat{x} + \mathcal{N}(A)$ for any particular solution $\hat{x}$.

Rank: $\text{rank}(A) = \dim \mathcal{R}(A) = \dim \mathcal{R}(A^*)$. Row rank = column rank.

Proof that row rank $\leq$ column rank: Choose a basis $v_1, \dots, v_{r_R}$ for the row space. Map to $Av_1, \dots, Av_{r_R}$ in the column space. These are linearly independent (if $\sum \alpha_i Av_i = 0$, then $A(\sum \alpha_i v_i) = 0$, so $w = \sum \alpha_i v_i \in \mathcal{N}(A) \cap \mathcal{R}(A^*) = \{0\}$, giving $\alpha_i = 0$). So column rank $\geq$ row rank; reverse by transpose.

CHAPTER 5: FACTORIZATIONS AND BASIS CHANGE (L07, L08, L09)

Simple Systems

Solving $Ax = b$ is easy when A is: - **Diagonal:** decouple equations, $x_i = b_i/a_{ii}$. - **Triangular:** forward/back substitution. - **Unitary/Orthogonal:** $x = A^{-1}b = A^*b$.

Strategy: Convert arbitrary A into a product of simple matrices.

Basis Change

In a basis given by columns of invertible T : the representation of the linear map A changes to $T^{-1}AT$.

Derivation: $x = T\tilde{x}$ (old-to-new), so $Ax = AT\tilde{x}$. The output in new coordinates is $T^{-1}ATx$.

Similar matrices ($B = T^{-1}AT$) have the same **eigenvalues**.

Eigenvalues and Diagonalization

Eigenvalue equation: $Ax = \lambda x$, $x \neq 0$.

Eigenspace: $\mathcal{N}(A - \lambda I)$.

Characteristic polynomial: $p(\lambda) = \det(\lambda I - A)$. Eigenvalues = roots.

Diagonalizable: $A = T\Lambda T^{-1}$ iff A has n linearly independent eigenvectors (columns of T).

Not all matrices are diagonalizable. But all square complex matrices can be **triangularized** (Schur).

Gram-Schmidt and Orthogonal Projections

Projection of x onto y : $\hat{x} = \frac{\langle x, y \rangle}{\|y\|^2} y = \frac{y^* x}{y^* y} y$. Error $x - \hat{x} \perp y$.

Projection onto subspace with ONB Q : $p = QQ^*x$. Projection matrix $P = QQ^*$.

Properties: $P^2 = P$ (idempotent), $P^* = P$ (self-adjoint), $\mathcal{R}(P) = \text{col space of } Q$.

For non-orthogonal columns A : $P = A(A^*A)^{-1}A^*$.

CHAPTER 6: UNITARY AND HERMITIAN MATRICES (L10, L11)

Unitary Matrices

$U^*U = UU^* = I$. Real case: orthogonal Q , $Q^T Q = I$.

Equivalent conditions: columns form ONB; rows form ONB; $U^{-1} = U^*$.

Norm preservation: $\|Ux\|_2 = \|x\|_2$. Proof: $\|Ux\|^2 = x^* U^* U x = x^* x$.

Inner product preservation: $\langle Ux, Uy \rangle = \langle x, y \rangle$.

Energy preservation: $\|UA\|_F = \|A\|_F$.

Eigenvalues on unit circle: If $Ux = \lambda x$, then $|\lambda| = 1$.

Proof: $\|Ux\|_2 = \|\lambda x\|_2 = |\lambda| \|x\|_2 = \|x\|_2$, so $|\lambda| = 1$.

Eigenspaces for distinct eigenvalues are orthogonal.

Hermitian Matrices

$A^* = A$. Real case: symmetric $A^T = A$.

Quadratic form: $x^* A x \in \mathbb{R}$ for all x .

Proof: $(x^* A x)^* = x^* A^* x = x^* A x$, so it equals its conjugate $\rightarrow$ real.

Eigenvalues real: From $x^* A x = \lambda x^* x$, since LHS is real and $x^* x > 0$, we get $\lambda \in \mathbb{R}$.

Eigenspaces for distinct eigenvalues orthogonal: Compare $\lambda\langle y, x \rangle = \langle Ax, y \rangle = \langle x, Ay \rangle = \mu\langle y, x \rangle^*$. Since λ, μ real and distinct: $\langle y, x \rangle = 0$.

Schur Factorization Theorem

Every square complex matrix has $A = UTU^*$ with U unitary, T upper triangular.

Proof idea: 1. Find an eigenvector u_1 (exists by fundamental theorem of algebra). 2. Extend $\{u_1\}$ to an orthonormal basis of $\mathbb{C}^n$. 3. In this basis, A has block upper triangular form (first column has only one nonzero entry). 4. Apply the same argument recursively to the $(n-1) \times (n-1)$ lower-right block.

CHAPTER 7: NORMAL MATRICES AND SPECTRAL THEORY (L11, L12, L13)

Normal Matrices

$$A^*A = AA^*.$$

Spectral theorem: A is normal $\iff A$ is unitarily diagonalizable: $A = U\Lambda U^*$.

Proof: Schur gives $A = UTU^*$. If A is normal, then T is normal. An upper triangular normal matrix must be diagonal. Conversely, if $A = U\Lambda U^*$, then $A^*A = U|\Lambda|^2U^* = AA^*$.

Energy identity: For normal A : $\|A\|_F^2 = \sum_i |\lambda_i|^2$. (Frobenius energy = eigenvalue energy.)

Subclasses of normal matrices:

Condition	Matrix Class	Eigenvalues
$A^* = A$	Hermitian	Real
$U^*U = I$	Unitary	On unit circle
$A^* = -A$	Skew-Hermitian	Imaginary

Positive Definite Matrices

For Hermitian A :

$$A \succ 0 \iff \text{all } \lambda_i > 0 \iff x^*Ax > 0 \forall x \neq 0 \iff A = SS^* \text{ for invertible } S.$$

Sign classes:

Class	Eigenvalues	Quadratic form shape
PD	All > 0	Convex paraboloid
PSD	All ≥ 0	Flat in zero-eigenvalue directions
ND	All < 0	Concave paraboloid
Indefinite	Mixed	Saddle

Diagonal dominance (sufficient, not necessary): $a_{ii} > \sum_{j \neq i} |a_{ij}|$ implies PD.

Positive linear combinations: If $A_i \succ 0$ and $c_i > 0$, then $\sum c_i A_i \succ 0$.

Submatrix property: If $A \succ 0$, then every principal submatrix (same row/column indices) is PD.

CHAPTER 8: INERTIA, STAR-CONGRUENCE, CHOLESKY (L13, L14, L15)

Inertia and Sylvester's Law

Inertia: $(n_+, n_-, n_0) = (\# \text{ positive}, \# \text{ negative}, \# \text{ zero eigenvalues})$, counted with multiplicity.

Star-congruence: $A \stackrel{*}{\sim} B$ means $A = SBS^*$ for invertible S .

Sylvester's law: $A \stackrel{*}{\sim} B \iff \text{inertia}(A) = \text{inertia}(B)$.

Similarity vs. star-congruence: - Similarity preserves **eigenvalues**. - Star-congruence preserves only **inertia** (sign pattern).

Corollary: $A \succ 0 \iff A \stackrel{*}{\sim} I$ (same inertia $(n, 0, 0)$) $\iff A = SS^*$.

Matrix Square Roots

For any invertible S with $SS^* = A$, S is a square root of A . There are infinitely many (insert any unitary factor T : $(ST)(ST)^* = STT^*S^* = SS^* = A$).

Positive definite square root: $A^{1/2} = U\Lambda^{1/2}U^*$ where $A = U\Lambda U^*$. Unique PD symmetric square root.

Cholesky Factorization

$A \succ 0 \implies \exists$ unique lower triangular L with positive diagonal: $A = LL^*$.

Proof sketch: Partition $A = \begin{bmatrix} \alpha & v^* \\ v & M \end{bmatrix}$. Star-congruence by block Gaussian elimination gives block diagonal $\begin{bmatrix} \alpha & 0 \\ 0 & M - vv^*/\alpha \end{bmatrix}$, which is PD by inertia preservation. The Schur complement $M - vv^*/\alpha$ is PD (not by direct computation — by star-congruence). Induct on dimension.

Application: Solve $Px = y$ (PD P) via $Lz = y$, then $L^*x = z$.

Least Squares Problem

For tall full-rank A ($m > n$, rank n) and general b :

$$\min_x \|Ax - b\|_2 \implies A^*A\hat{x} = A^*b \implies \hat{x} = (A^*A)^{-1}A^*b.$$

Normal equations from orthogonality: $b - A\hat{x} \perp \mathcal{R}(A)$.

$A^*A \succ 0$ when A has full column rank: proof from $\|Ax\|^2 = x^*(A^*A)x \geq 0$ with equality only for $x = 0$.

Numerically: Use QR or SVD; avoid forming A^*A .

Random Vectors: Covariance and Square Roots

Covariance matrix: $C_x = \mathbb{E}[(x - \mu_x)(x - \mu_x)^*]$. Always Hermitian and PSD; PD if full rank.

Coloring: $y = C_y^{1/2}z$ where z has identity covariance $\rightarrow y$ has covariance C_y .

Whitening: $q = C_y^{-1/2}y \rightarrow q$ has identity covariance.

CHAPTER 9: QR FACTORIZATION (L16, L17)

QR Factorization

Any $m \times n$ matrix A (with $m \geq n$) can be written:

$$A = QR,$$

where Q has orthonormal columns and R is upper triangular.

- **Reduced QR:** Q is $m \times n$, R is $n \times n$.
- **Full QR:** Q is $m \times m$ unitary, R is $m \times n$.

Gram-Schmidt Process

Build orthonormal $q_1, \dots, q_n$ from columns $a_1, \dots, a_n$ of A :

$$q_k = \frac{a_k - \sum_{i < k} (q_i^* a_k) q_i}{\|a_k - \sum_{i < k} (q_i^* a_k) q_i\|}$$

Causal constraint: $\text{span}\{q_1, \dots, q_k\} = \text{span}\{a_1, \dots, a_k\}$ for all $k \rightarrow$ upper triangular R .

Linear dependence: $\tilde{a}_k = 0 \rightarrow$ no new direction; rank of $A < n$.

Modified Gram-Schmidt: Orthogonalize against each q_k immediately after computing it; numerically more stable.

Circulant Matrices and Fourier Diagonalization (L16)

Circulant matrix: each row is a circular shift of the previous.

DFT eigenvectors: Complex exponentials $f_k = (1, e^{j2\pi k/n}, \dots, e^{j2\pi k(n-1)/n})^T$ are eigenvectors:

$$Hf_k = \hat{h}_k f_k,$$

where $\hat{h}_k$ is the DFT of the impulse response at frequency k .

DFT matrix: $F = [f_0 \ f_1 \ \dots \ f_{n-1}]$. $F^*F = nI$ (not unitary, but scaled unitary).

Diagonalization: $H = F\Lambda F^{-1} = \tilde{F}\Lambda\tilde{F}^*$ where $\tilde{F} = F/\sqrt{n}$ is unitary.

Consequence: All circulant matrices are normal matrices; convolution in time = multiplication in frequency.

FFT: Computes $\tilde{F}^*x$ in $O(n \log n)$ instead of $O(n^2)$.

CHAPTER 10: BLOCK MATRIX TECHNIQUES (L18)

Block LDU Factorization

For $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ with A_{11} invertible:

$$A = \begin{bmatrix} I & 0 \\ A_{21}A_{11}^{-1} & I \end{bmatrix} \begin{bmatrix} A_{11} & 0 \\ 0 & S_{11} \end{bmatrix} \begin{bmatrix} I & A_{11}^{-1}A_{12} \\ 0 & I \end{bmatrix},$$

where $S_{11} = A_{22} - A_{21}A_{11}^{-1}A_{12}$ is the **Schur complement** of A_{11} .

Determinant: $\det(A) = \det(A_{11}) \det(S_{11})$.

PD criterion (Hermitian case): $A \succ 0 \iff A_{11} \succ 0$ and $S_{11} \succ 0$.

Inertia additivity (Hermitian): $\text{inertia}(A) = \text{inertia}(A_{11}) + \text{inertia}(S_{11})$.

Matrix Inversion Lemma (Woodbury Identity)

$$(A + BCD)^{-1} = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1}.$$

Application: Rank-one update of matrix inverse; used in Recursive Least Squares (RLS).

CHAPTER 11: MATRIX NORMS (L19, L20)

Frobenius Norm

$$\|A\|_F = \sqrt{\text{tr}(A^*A)} = \sqrt{\sum_k \sigma_k^2}.$$

Induced Norms

$$\|A\|_{q,p} = \sup_{\|x\|_p=1} \|Ax\|_q.$$

Shortcuts: - $\|A\|_{1,1} = \max_j \|a_j\|_1$ (max column 1-norm). - $\|A\|_{\infty,\infty} = \max_i \|r_i\|_1$ (max row 1-norm).
- $\|A\|_{2,2} = \sigma_1(A)$ (largest singular value = operator norm).

Geometric interpretation: Unit ℓ_2 sphere maps to ellipsoid; longest semi-axis = σ_1 .

Schatten p -Norms

$$\|A\|_{S_p} = \|\sigma\|_p = \left(\sum_k \sigma_k^p \right)^{1/p}.$$

- $p = 1$: nuclear norm $\|A\|_* = \sum_k \sigma_k$.
- $p = 2$: Frobenius norm.

- $p = \infty$: operator norm σ_1 .

Nuclear norm: Convex relaxation of rank minimization. Used in matrix completion (e.g., Netflix problem).

Weighted 2-Norm

$\|x\|_W = \sqrt{x^* W x}$ for PD W . Eigenvectors of W define weighted directions; eigenvalues are weights. Appears naturally in multivariate Gaussian: the precision matrix Σ^{-1} is the weighting.

CHAPTER 12: SINGULAR VALUE DECOMPOSITION (L21, L22)

Definition

For any $m \times n$ matrix A of rank r :

$$A = U \Sigma V^*, \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0.$$

- U : $m \times m$ unitary (columns = left singular vectors u_k).
- V : $n \times n$ unitary (columns = right singular vectors v_k).
- Σ : $m \times n$ real nonneg diagonal.

Outer product: $A = \sum_{k=1}^r \sigma_k u_k v_k^*$ (rank-1 sum, terms mutually orthogonal).

Geometric Foundation

Unit sphere in $\mathbb{C}^n$ maps to an ellipsoid in $\mathbb{C}^m$. Principal semi-axis lengths = singular values. Semi-axis directions = left singular vectors. Pre-images = right singular vectors.

Singular Values: Properties

- Always real and nonneg (they are norms of images of unit vectors).
- σ_k^2 = eigenvalues of $A^* A$ (= eigenvalues of AA^*).
- Right singular vectors = eigenvectors of $A^* A$.
- Left singular vectors = eigenvectors of AA^* .

Full and Reduced SVD

- **Full SVD:** U is $m \times m$, V is $n \times n$.
- **Reduced (compact) SVD:** $\hat{U}$ is $m \times r$, $\hat{\Sigma}$ is $r \times r$, $\hat{V}$ is $n \times r$.

SVD and Four Fundamental Subspaces

Subspace	Basis
$\mathcal{R}(A)$	Columns 1 to r of U
$\mathcal{N}(A^*)$	Columns $r + 1$ to m of U
$\mathcal{R}(A^*)$	Columns 1 to r of V
$\mathcal{N}(A)$	Columns $r + 1$ to n of V

Norms from SVD

$$\|A\|_2 = \sigma_1, \quad \|A\|_F = \sqrt{\sum_{k=1}^r \sigma_k^2}, \quad \text{rank}(A) = r.$$

Best Rank- p Approximation (Eckart-Young)

$$A_p = \sum_{k=1}^p \sigma_k u_k v_k^*, \quad \|A - A_p\|_F = \sqrt{\sum_{k=p+1}^r \sigma_k^2}.$$

A_p is the closest rank- p matrix in both Frobenius and 2-2 norms.

CHAPTER 13: ABSTRACT INNER PRODUCT SPACES (L23)

Inner Product Space Axioms

Function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$: 1. Conjugate symmetry: $\langle x, y \rangle = \overline{\langle y, x \rangle}$. 2. Linear in first argument. 3. Positive definite: $\langle x, x \rangle \geq 0$; $= 0 \iff x = 0$.

Induced norm: $\|x\| = \sqrt{\langle x, x \rangle}$. All geometric concepts (orthogonality, projection, Cauchy-Schwarz) generalize.

Hilbert space: complete inner product space.

Examples

- $\mathbb{C}^n$: $\langle x, y \rangle = y^* x$. Induces Euclidean norm.
- $L^2([a, b])$: $\langle f, g \rangle = \int_a^b f(t) \overline{g(t)} dt$. Induces L^2 norm.
- **Matrices:** $\langle A, B \rangle = \text{tr}(A^* B)$. Induces Frobenius norm. SVD outer products are an orthonormal basis.

Why Inner Products Enable Closed Forms

Optimization problems involving the induced norm have closed-form solutions via the orthogonality principle: the optimal error is orthogonal to the constraint set. This generalizes least squares, projection, Fourier series coefficients.

Nuclear Norm and Matrix Completion (L23)

Matrix completion (Netflix): Fill in unknown entries of a matrix using the low-rank hypothesis.

Nuclear norm minimization: $\min_B \|B\|_*$ s.t. $B_{ij} = A_{ij}$ for known entries. Convex; recovers the true low-rank matrix when enough entries are observed (under incoherence conditions).

KEY THEOREMS SUMMARY

Theorem	Statement
Solvability	$Ax = b$ has solution iff $b \in \mathcal{R}(A)$
Solution set	All solutions: $\hat{x} + \mathcal{N}(A)$
Null $\perp$ Row	$\mathcal{N}(A) \perp \mathcal{R}(A^*)$
Rank	Row rank = column rank
Cauchy-Schwarz	$ \langle x, y \rangle \leq \ x\ \ y\ $
Schur	Every square complex $A = UTU^*$ (U unitary, T triangular)
Spectral	A normal $\iff A = U\Lambda U^*$
Sylvester	$A \stackrel{*}{\sim} B \iff \text{inertia}(A) = \text{inertia}(B)$
Cholesky	$A \succ 0 \iff A = LL^*$ (lower triangular, positive diagonal)
SVD	Any $A = U\Sigma V^*$ (singular values real, nonneg)
Eckart-Young	$A_p = \sum_{k=1}^p \sigma_k u_k v_k^* = \text{best rank-}p \text{ approx}$
Woodbury	$(A + BCD)^{-1} = A^{-1} - A^{-1}B(\dots)^{-1}DA^{-1}$